Problem Set 3: Question 5 and 6

Econ 508: Applied Econometrics

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October 23, 2013

Question 5 and 6: Objectives

The main objective is to look at the dynamic behavior of the cobweb model:

$$Q_t = \alpha_1 + \alpha_2 p_{t-1} + \alpha_3 z_t + u_t \tag{1}$$

$$p_t = \beta_1 + \beta_2 Q_t + \beta_3 w_t + v_t, \tag{2}$$

Dynamics

1. Quantity supplied Q_t depends on the previous period's price p_{t-1} arguing that in the world producers decide how much to offer based on the last observed price plus and random error u_t.
2. This production decision at time t is used to determine the market clearing price in demand. In simple words, last periods price determines current period supply, whereas current period demand determines the market clearing price in the market.

Summary statistics of the data

	p_t	Q_t	z_t	w_t
Min.	1.200	-1.026	0.3156	0.4373
1st Qu.	7.474	7.328	1.1087	1.0286
Median	10.060	9.743	1.4744	1.5545
Mean	10.146	10.095	1.8702	1.5682
3rd Qu.	12.794	13.105	2.6213	1.9673
Max.	25.695	25.319	4.8036	3.4797

OLS Estimates: Supply

With the assumption that u_t and v_t are, we are able to estimate the model recursively by the usual OLS approach, since p_{t-1} is exogenous in this setting.

Table : OLS Estimates – Supply

	Estimate	Std. Error	t value	Pr(>|t|)
Intercept	-0.3872	0.6114	-0.63	0.5273
p_{t-1}	0.9422	0.0441	21.36	0.0000***
z_t	0.4860	0.1735	2.80	0.0056**

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OLS Estimates: Demand

Table : OLS Estimates – Demand

	Estimate	Std. Error	t value	Pr(>|t|)
Intercept	8.5854	0.6386	13.44	0.0000
Q_t	-0.4854	0.0497	-9.76	0.0000
w_t	4.1194	0.3521	11.70	0.0000

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Dynamic Behavior

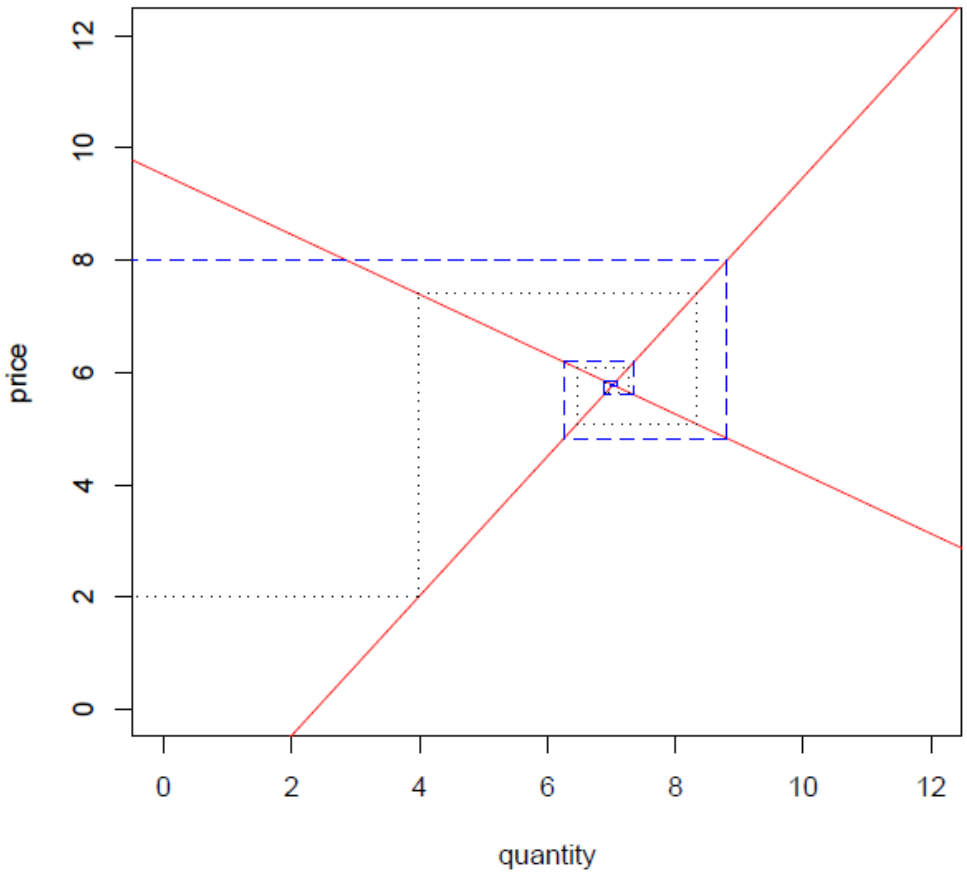

Figure : Dynamic Behavior of Market Equilibrium

Forecasts

To make forecasts we begin assuming z_t and w_t are fixed at their last observed values, z_T and w_T. Then,in equilibrium $p_{t-1} = p_t$, so the equilibrium quantity and price are given by

$$Q_e = \frac{(\alpha_1 + \alpha_2\beta_1 + \alpha_2\beta_3 w_T + \alpha_3 z_T)}{(1 - \alpha_2\beta_2)} \tag{3}$$

$$p_e = \frac{(\beta_1 + \alpha_1\beta_2 + \alpha_3\beta_2 z_T + \beta_3 w_T)}{(1 - \alpha_2\beta_2)} \tag{4}$$

Therefore, based on the estimated coefficients, we find that $\hat{Q}_e = 8.137521$ and $\hat{p}_e = 8.786806$.

Forecasts Quantity

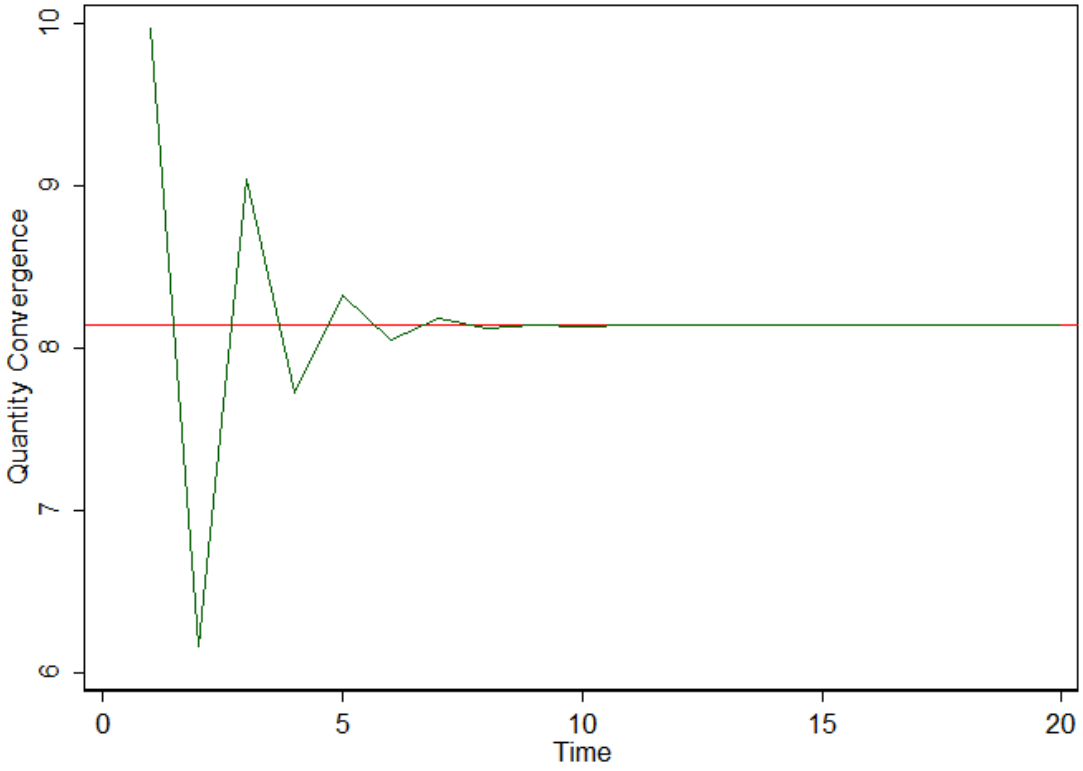

Figure : Quantity Convergence

Forecasts Price

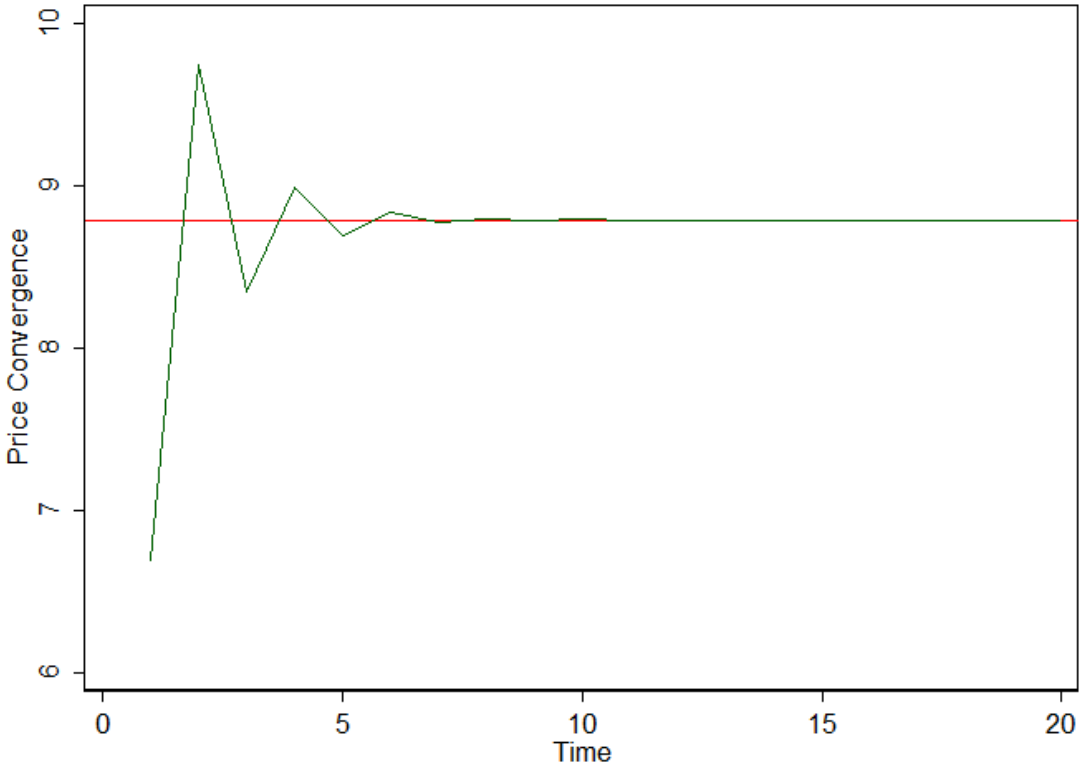

Figure : Price Convergence

Final Issues

Table : Specification Tests

		Supply		Demand	
Test	H_0	T	p-value	T	p-value
RESET	Linear Cond. Mean	0.0577	0.9818	0.0924	0.9641
Shapiro-Wilk	Normally distributed residuals	-0.3897	0.6516	0.4965	0.3098
Breusch-Pagan	Homoskedasticity *(under normality)*	0.0141	0.9054	0.0000	0.9973

Then testing for autocorrelation would be the next natural step.

Thanks